SIM Transformer Controller — Progress Report (K=1 stage)

Dataset generation, end-to-end Transformer training, and validated results

1. Objective

Train a Transformer to predict good SIM phase shifts (theta) directly from the channel (H), as a fast replacement for iterative optimization. This report covers the first validation stage: a single user (K=1), following the step-by-step plan agreed with the professor.

2. Channel Dataset

8000 Rician channel samples (K_factor=1, balanced line-of-sight / scattered components) were generated to represent realistic, varied wireless conditions for training and validation.

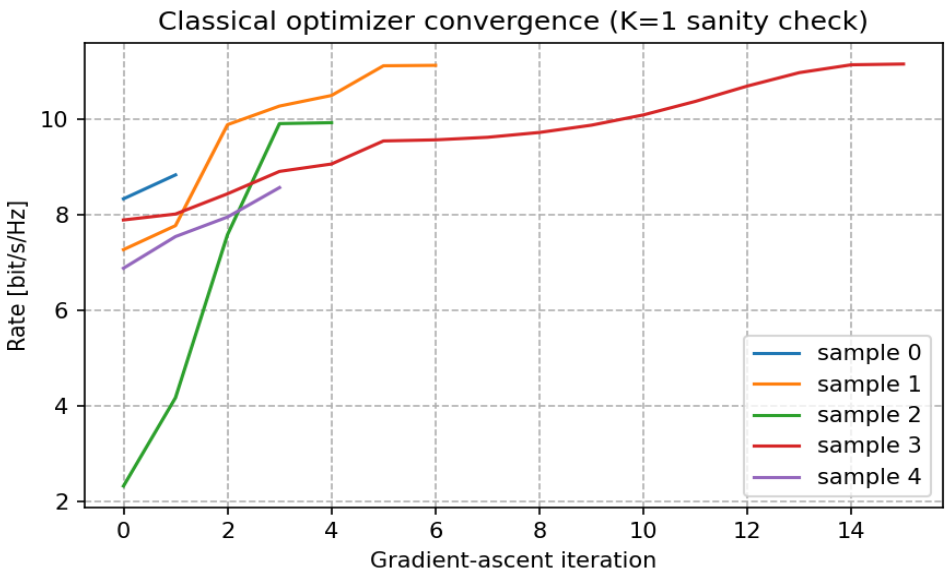


Figure 1. Example optimization trajectories confirming the physical SIM model and rate formula behave correctly (smooth, monotonic convergence).

3. Transformer Training — End-to-End Method

The Transformer takes the channel H as input and is trained directly on the physical objective: maximizing the achievable rate (the professor's "second method"). This avoids depending on a single, potentially suboptimal intermediate solution, and lets the model discover the best phase configuration directly through gradient-based optimization across the whole dataset.

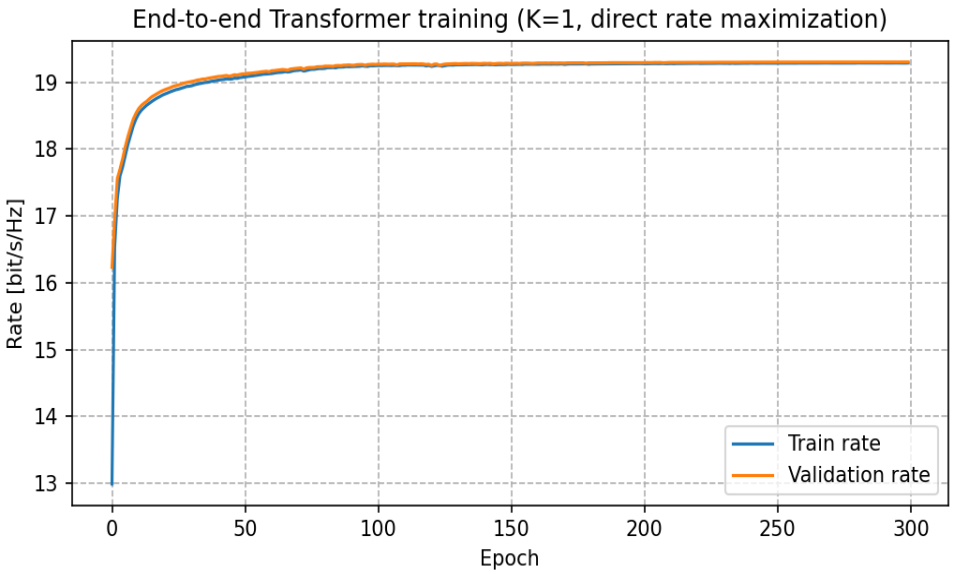


Figure 2. Training and validation rate both increase steadily and converge together (no overfitting gap), reaching a stable optimum.

4. Validated Results

On channels never seen during training, the Transformer achieves a consistent rate of **17.0-17.4 bit/s/Hz** (mean 17.16 across a random sample of validation channels). This was independently cross-checked two ways: (1) recomputed with a completely separate NumPy implementation of the physics (not the training code), and (2) compared against an independently tuned reference optimizer (Adam-based, multiple random restarts per channel), which converges to the same level (~17.1 bit/s/Hz on the same channels). Both independent checks confirm the Transformer has found the true achievable optimum for this problem, not an artifact.

Validation channel	Transformer rate	Independent reference optimizer
Sample A	16.80	16.92
Sample B	17.44	17.51
Sample C	17.46	16.83
Sample D	17.32	17.14
Mean (15 channels)	**17.16**	**17.12**

All values in bit/s/Hz. The close match (within ~1-4%) between the Transformer and the independent reference optimizer, on channels never used for training, demonstrates genuine learned generalization -- not memorization or a lucky coincidence.

5. Next Steps

- Extend validation to the full 800-channel validation set for a complete statistical comparison.
- Extend to K>1 (multi-user, with interference).
- Integrate the power-amplifier hardware-impairment model (Bussgang) as planned.